

Minimodule 3. Induction

List of exercises (from 7th edition of the book)

Section 5.1 Exercises 3, 9, 19, 31

3. Let $P(n)$ be the statement that $1^2 + 2^2 + \cdots + n^2 = n(n+1)(2n+1)/6$ for the positive integer n .
- What is the statement $P(1)$?
 - Show that $P(1)$ is true, completing the basis step of the proof.
 - What is the inductive hypothesis?
 - What do you need to prove in the inductive step?
 - Complete the inductive step, identifying where you use the inductive hypothesis.
 - Explain why these steps show that this formula is true whenever n is a positive integer.

9. a) Find a formula for the sum of the first n even positive integers.
b) Prove the formula that you conjectured in part (a).

19. Let $P(n)$ be the statement that

$$1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n},$$

where n is an integer greater than 1.

- What is the statement $P(2)$?
- Show that $P(2)$ is true, completing the basis step of the proof.
- What is the inductive hypothesis?
- What do you need to prove in the inductive step?
- Complete the inductive step.
- Explain why these steps show that this inequality is true whenever n is an integer greater than 1.

31. Prove that 2 divides $n^2 + n$ whenever n is a positive integer.

For more practice, you may also consider the following exercises 32 - 37, which are similar but require some other rearrangements to apply the inductive hypothesis.

Apply the steps in the inductive proof - basis step $P(1)$, identity the I.H., inductive step $P(k) \rightarrow P(k+1)$ and what to prove; finally, complete the proof using the I.H. and conclude.

Note: There are many ways to proceed, but start from the "left side" of what to prove, rewrite to apply the I.H. before manipulating, then look towards what you need to conclude in the end - can you simplify already from the start?

The sum over $2i$, with i from 1 to n equals $n(n+1)$. Prove as above for all natural numbers.

Same procedure, start from the left and apply the I.H.. Try to isolate the terms so that "it looks like" what you need to prove, and once you are there, prove that the left-over term only makes the result smaller (remember, it is an inequality); alternatively, one can do a shortcut by proving that what's left on the "left side" after applying the I.H. is less than the corresponding term on the "right side", that is, in this case use both sides (not generally recommended) to conclude that $-1 < 0$.

If 2 divides $n^2 + 2$ then $n^2 + n = 2a$, where a belongs to the set of positive integers; use this fact as the I.H. to conduct the inductive proof.